

A cost-effective & repeatable method for characterisation of functionalised carbon nano-materials

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Abstract

Although graphene has been shown to exhibit numerous exciting and supreme properties, there remain many challenges facing the material if it is to be truly commercialised. Two key, but independent, problems for the emerging graphene industry are:

(1) The requirement of chemical functionalisation for real-world applications (essential for homogenous dispersion and application); and

(2) The pressing need for reliable, cost-effective, rapid, easy-to-use and accurate characterisation techniques to determine the type and extent of functionalisation present on graphene material.

Current chemical characterisation requires a high level of competency and expert ability to interpret results. X-ray photoelectron spectroscopy (XPS), Raman and Fourier Transform Infrared (FTIR) spectroscopy are costly, time-consuming and typically not available to the host of UK SMEs without sub-contracting. Thus quality control present in the production and functionalisation of graphene is currently very limited.

Haydale has developed an 'on-site' and 'rapid' characterisation method to confirm functionalisation of graphene material.

Introduction

Functionalisation refers to the addition of chemical groups to sites on the surface of carbon nanoparticles. This is done to enhance the compatibility between the nanoparticles and the bulk matrix in which the particles are dispersed, figure 1. Surface functionalisation of carbon nanoparticles is historically achieved through wet acid processing [1]; however, Haydale's patent-pending, environmentally friendly HDPlas™ plasma processing [2] method has recently received a lot of attention [3, 4].

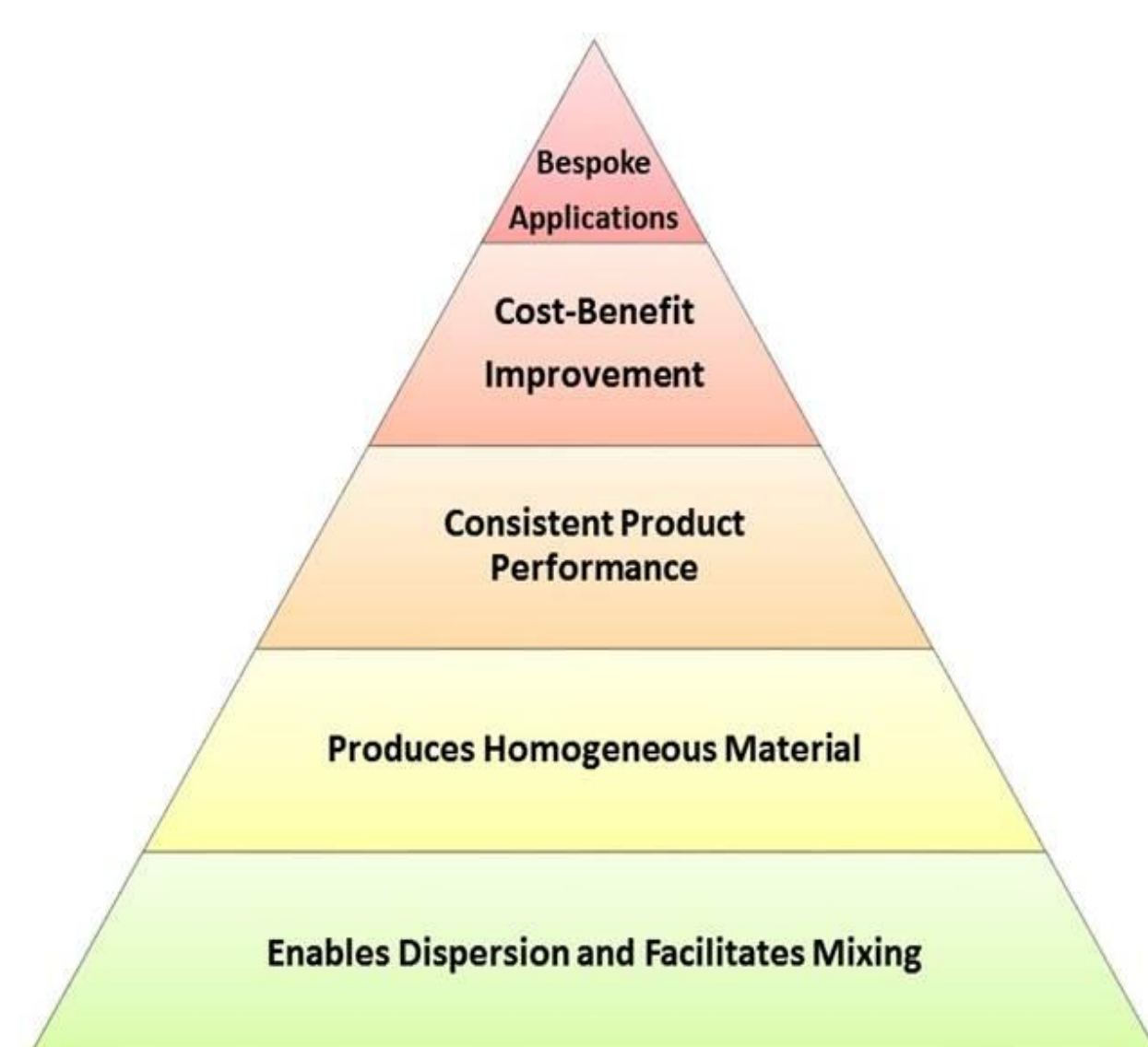


Figure 1 Advantages of HDPlas™ plasma functionalised carbon nanomaterials in composite systems.

Most particles, including surface functionalised carbon nanoparticles, dispersed in an aqueous system acquire a surface charge, principally either by ionisation of surface groups, or adsorption of charged species. These surface charges modify the distribution of the surrounding ions, resulting in a layer around the particle that is different to the bulk solution. The stability of a given dispersion depends on the particles' zeta potential [5].

It is known that surface functionalisation of nanoparticles can improve their dispersion in water and other liquids [6] by modifying the particles' zeta potential. The stabilising effect of surface functionalisation is easily verified by observation of the dispersion stability over a period of time, figure 2.

Experimental

The dispersion analysis tool, figure 3, consists of a white light emitting diode (LED) and a photodiode. Light transmitted through the graphene (GNP)-liquid dispersion is detected by the photodiode. Light energy is converted to a voltage by monitoring from a high input impedance device.



Figure 2 Dispersion of graphene nanoplatelets (GNPs) in deionised water.



Figure 3 Haydale's 4-channel 'Dispersion Stability Analyser'.

The main variables that determine dispersion stability are:

Surface functionalisation; viscosity of the liquid; solid:liquid ratio; and consistency of dispersion.

In this work, we test the dispersion stability of the following carbon nanomaterials in deionised water: Pristine ('as-received') GNPs; GNP-COOH; GNP-NH₃; GNP-N₂; GNP-O₂; and GNP-F.

Results

Dispersion stability of plasma functionalised GNPs is typically referenced to pristine (non-plasma processed) GNPs. The dispersion stability of various surface functionalisations is shown in figure 4. It can be seen that the various functionalisations exhibit different stabilities. Fluorine functionalisation is quite unstable, but not as unstable as pristine GNPs. At the opposite end we confirm that oxygen containing functional groups produce the most stable GNP dispersions in water.

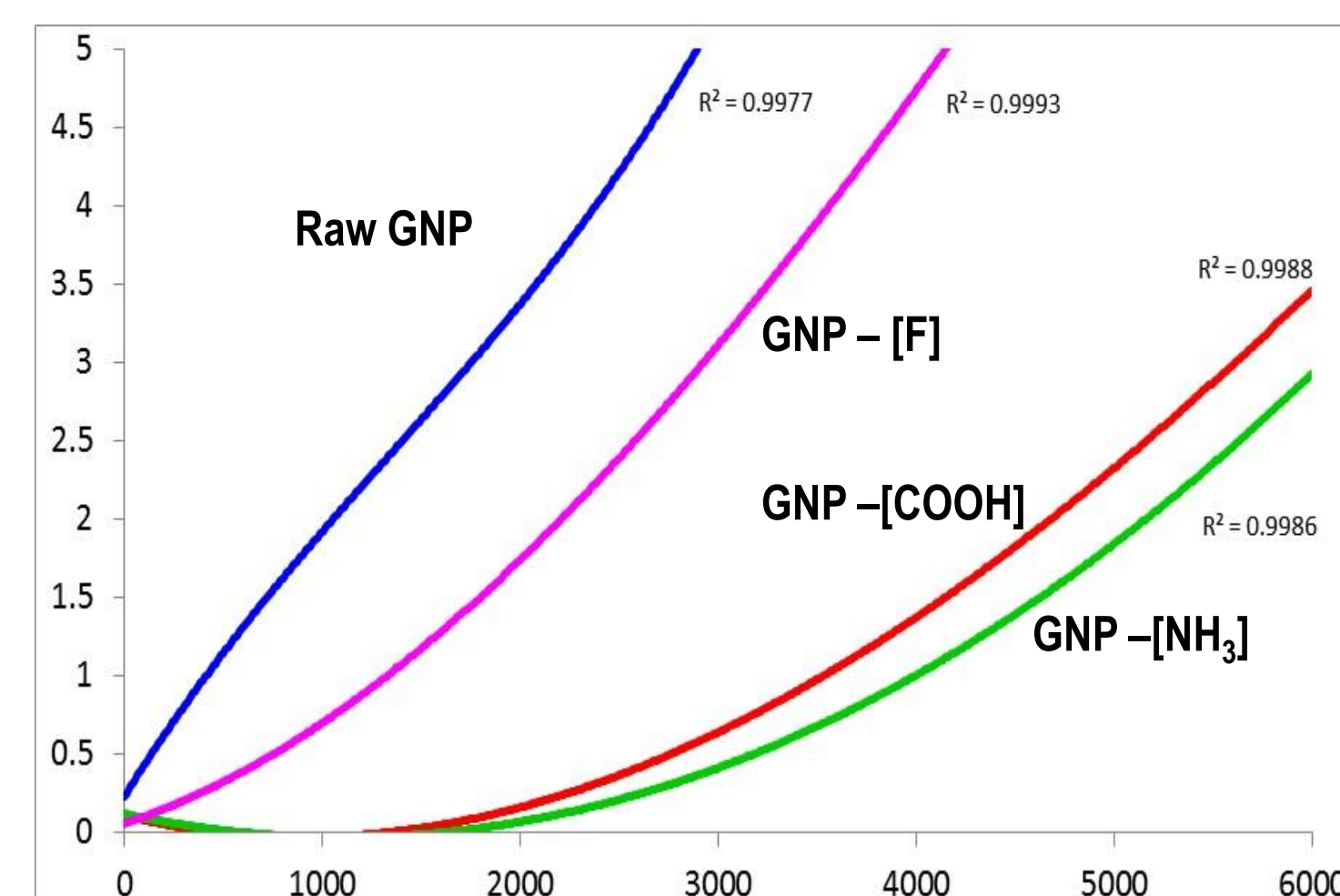


Figure 4 Dispersion stability: There is a clear difference between dispersion curves of materials with various different surface functionalisations.

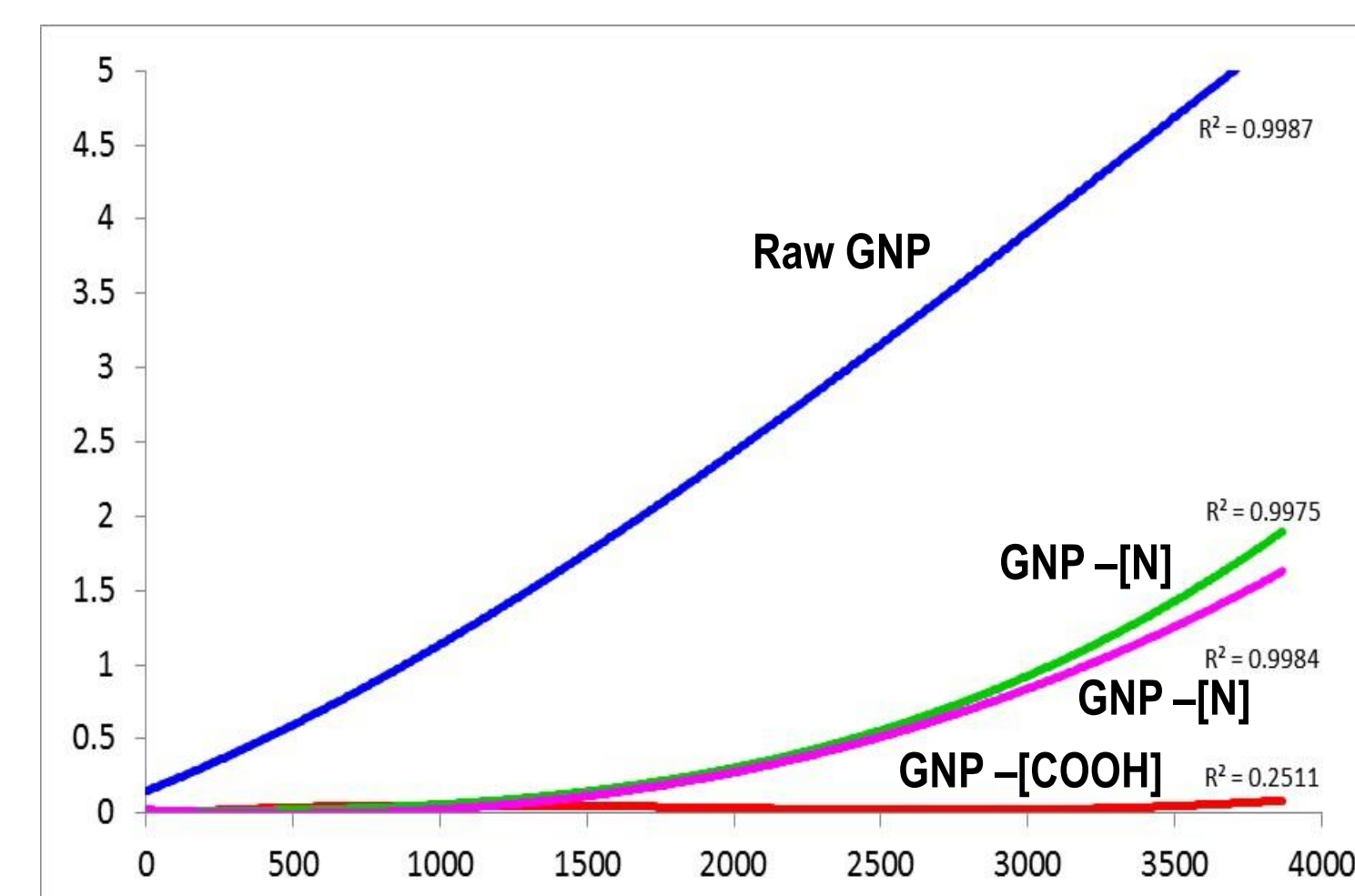


Figure 5 Dispersion curves can be used to show that (i) plasma functionalisation has taken place and (ii) the most likely functional gases used.

Testing multiple samples in parallel has the advantage of allowing us to compare samples with known reference samples, thereby allowing us to improve repeatability, see figure 5.

Another important feature of our test is its ability to distinguish between different levels of functionalisation; we see an example of this in figure 6. Increasing oxygen functionalisation (as measured by XPS) gives more stable dispersions in water. This level of sensitivity is difficult, if not impossible, to achieve using other, more traditional analytical techniques.

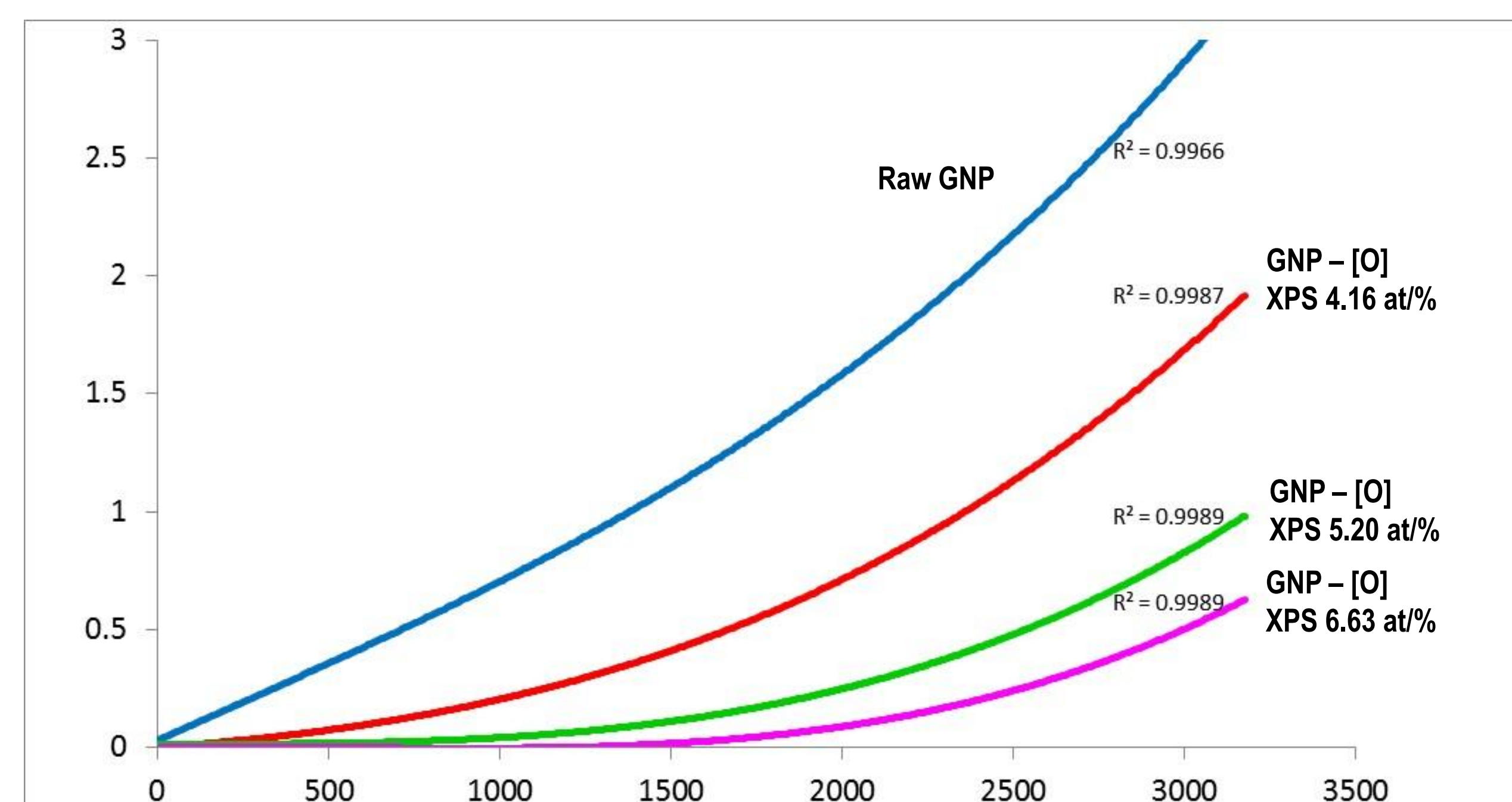


Figure 6 Dispersion stability of oxygen functionalised GNPs, showing that the tool can discriminate between various levels of functionalisation.

Summary

Haydale's dispersion stability test provides:

- A rapid, simple and repeatable test to confirm the effectiveness of the plasma functionalisation process;
- A quality control technique to complement traditional analysis methods currently used by industry;
- The ability to indicate the level of functional groups added;
- A commercially viable and attractive process control tool for manufacturing; and
- An ability to discriminate between different functional groups.

References

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